



NSW Education Standards Authority

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Centre Number

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Student Number

2023 HIGHER SCHOOL CERTIFICATE EXAMINATION

Food Technology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Write your Centre Number and Student Number at the top of this page

Total marks:
100

Section I – 20 marks (pages 2–6)

- Attempt Questions 1–20
- Allow about 35 minutes for this section

Section II – 50 marks (pages 9–15)

- Attempt Questions 21–26
- Allow about 1 hour and 25 minutes for this section

Section III – 15 marks (page 33)

- Attempt Question 27
- Allow about 30 minutes for this section

Section IV – 15 marks (page 34)

- Attempt Question 28
- Allow about 30 minutes for this section

Section I

20 marks

Attempt Questions 1–20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1–20.

- 1** Vacuum sealing stops food from spoiling by controlling the
- A. light.
 - B. oxygen.
 - C. enzymes.
 - D. temperature.
- 2** The excess consumption of sugar is most likely to be responsible for causing which condition?
- A. Constipation
 - B. Bowel cancer
 - C. Osteoporosis
 - D. Dental caries
- 3** A food manufacturer uses semi-automated commercial equipment to produce baked goods.
- This is an example of which level of operation?
- A. Household
 - B. Multinational
 - C. Small business
 - D. Large company
- 4** Which of the following products would be suitable for an individual living with coeliac disease?
- A. Gluten-free pasta
 - B. High-fibre rye bread
 - C. Barley grain porridge
 - D. Wheat flour chocolate cake

- 5** Why would a food manufacturing company use the hazard analysis and critical control point (HACCP) system?
- A. To analyse staff productivity
 - B. To manage processing issues
 - C. To improve energy efficiency
 - D. To ensure consistent product size
- 6** Which of the following lists long-term physical effects of type 2 diabetes?
- A. Bloating and lethargy
 - B. Blindness and renal failure
 - C. Mood swings and mouth ulcers
 - D. Low-blood glucose and reduced thirst
- 7** What factor of the macro-environment could affect a food manufacturer?
- A. Change of government
 - B. The wider community's perception of the food manufacturer
 - C. Improvement to production facilities which increase efficiency and volume
 - D. The food manufacturer's financial ability to invest in research and development
- 8** Which of the following tasks in food manufacturing would be performed by unskilled workers?
- A. Conducting sensory assessments
 - B. Development of packaging labels
 - C. Visual inspection of fresh produce
 - D. Modification of processing equipment
- 9** Which of the following is an example of a food industry advisory group?
- A. NSW Food Authority
 - B. Food and Agriculture Organization
 - C. Australian Food and Grocery Council
 - D. Australian Institute of Science and Technology

10 Which of the following must appear on a food label in Australia?

- A. Nutrition panel
- B. Guaranteed health claims
- C. Directions for the disposal of the packaging
- D. Contact information of overseas manufacturer

11 A food item is recalled as it contains an undeclared allergen.

What are the next steps to be followed?

- A. Conduct a batch control test and donate recalled products to a charity
- B. Release a media statement and direct supermarkets to discount the product
- C. Notify consumers of the inaccuracy and modify ingredients label on all existing products
- D. Complete a food safety audit and advise consumers to return the product to place of purchase

12 Eating a diet high in soy will provide an individual with

- A. probiotics.
- B. carotenoids.
- C. insoluble fibre.
- D. phytoestrogens.

13 Which of the following is an example of a *me too* food product?

- A. Launching a generic breakfast cereal
- B. Using organic ingredients for a muffin range
- C. Packaging mini chocolate bars in bulk packets
- D. Producing a low sugar version of an existing muesli bar

- 14** An individual has a subscription to a food company that sells pre-packaged dinners.

Who is responsible for the nutritional impact of these food products on the individual?

- A. The consumer
- B. Private agencies
- C. The government
- D. The meal deliverer

- 15** Which of the following lists of additives is used to fortify foods?

- A. Folate, calcium, iron
- B. Vitamin C, gluten, sucrose
- C. Lactose, vitamin D, gelatin
- D. Dietary fibre, sodium, iodine

- 16** A food product developer conducted market research and it identified more people were choosing to eat a meat-free meal once a week.

What is the next step, in response to this research, for the food product developer?

- A. Create a marketing campaign that targets vegans
- B. Conduct idea generation for potential new food products
- C. Launch the prototype of a new food item into the marketplace
- D. Examine the break-even point of their existing vegetarian range

- 17** Which row of the table identifies a potential benefit and a potential risk when producing organic tomatoes?

	<i>Benefit</i>	<i>Risk</i>
A.	The yield will be consistent throughout the season	The tomatoes will all be the same size
B.	Pests can be managed by the use of companion planting	Some genes in the tomatoes have been modified
C.	The tomatoes will be free from synthetic chemicals	There will be a greater variety in the quality of the produce
D.	The tomatoes will be more nutritious than non-organic tomatoes	There are concerns about the safety of organically produced foods

18 The use of oxygen scavengers in the packaging of bread products is an example of

- A. gas packaging.
- B. active packaging.
- C. aseptic packaging.
- D. barrier specific packaging.

19 A food company has launched a new line of snacks targeting health-conscious consumers. They claim that their snacks are made with '100% natural ingredients' and are 'free from artificial additives'. The company's advertisements show images of fresh fruits and vegetables, giving the impression that their snacks are healthy and nutritious.

What is a possible ethical issue associated with the advertising claims made by the food company?

- A. Meeting the demands of the target market
- B. Exaggerating the taste and flavour of the snacks
- C. Misleading consumers about the nutritional value of the snacks
- D. Failing to provide contact information on the snack packaging label

20 What is the primary role of the Department of Agriculture, Fisheries and Forestry?

- A. Oversee quality control procedures and guarantee that food is safe for consumption
- B. Manage disease outbreaks and facilitate the movement of all stock around the country
- C. Inspect food production facilities and ensure that Australian produce is free of antibiotics
- D. Prevent the entry of exotic pests into Australia and certify livestock exports for the global market

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Centre Number

Food Technology

Sections II, III and IV

Answer Booklet

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Student Number

Instructions

- Write your Centre Number and Student Number at the top of this page.
- Use this answer booklet for Sections II, III and IV.
- Follow the instructions inside this booklet as to where you should write your answers.

Please turn over

Section II

50 marks

Attempt Questions 21–26

Allow about 1 hour and 25 minutes for this section

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (4 marks)

(a) Identify TWO foods that contain probiotics. 2

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(b) Describe the role probiotics play in a balanced diet. 2

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Question 22 (6 marks)

- (a) Outline ONE food safety hazard associated with selling uncovered raw meat in a chilled open display cabinet. **2**

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- (b) Describe a process for maintaining the quality of seafood products. **4**

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Question 23 (8 marks)

- (a) Explain why a food manufacturer may use a SWOT analysis in the development of a new product.

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- (b) Describe how chocolate-coated biscuits should be packaged.

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Question 24 (11 marks)

- (a) Describe TWO gender issues in ONE sector of the Australian food industry. **5**

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- (b) Explain how waste is managed in the food processing and manufacturing sector. Provide TWO relevant examples. **6**

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Question 25 (10 marks)

- (a) Outline the characteristics of ONE group at risk of malnutrition.

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- (b) Describe a diet-related condition that is experienced by a specific group in Australia.

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- (c) Explain ONE strategy that aims to promote optimal health for nutritionally vulnerable individuals.

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Question 26 (11 marks)

- (a) Compare TWO price structures a food company may choose to use. Include relevant examples.

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- (b) Explain THREE factors a food company must consider when determining a price for their product.

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Section III — Answer Question 27 on pages 16–22

27

Question Number

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If you require more space to answer Section III, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section III, tick here.

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Section IV — Answer Question 28 on pages 23–29

28

Question Number

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If you require more space to answer Section IV, you may ask for an extra writing booklet.

If you have used an extra writing booklet for Section IV, tick here.

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Food Technology

Section III

15 marks

Attempt Question 27

Allow about 30 minutes for this section

Answer the question on pages 16–22 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Question 27 (15 marks)

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|-----|--|----------|
| (a) | Outline factors that influence the availability of fresh produce in food retail. | 3 |
| (b) | Describe commercial practices that food companies could implement in response to consumer demands. | 4 |
| (c) | Explain preservation processes for extending the shelf life of ONE perishable food. | 8 |

Please turn over

Section IV

15 marks

Attempt Question 28

Allow about 30 minutes for this section

Answer the question on pages 23–29 of the Sections II, III and IV Answer Booklet. Extra writing booklets are available.

Your answer will be assessed on how well you:

- demonstrate knowledge and understanding relevant to the question
 - apply course concepts to food technology issues
 - communicate ideas and information using appropriate terminology and relevant examples
 - present a logical and cohesive response
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Question 28 (15 marks)

Discuss the impact of government policies and legislation on the agri-food chain.

End of paper